

Array yard

Revise 360 lesson plan — 60 minutes

Unit	OCR J277 2.2 Programming fundamentals — Paper 2, Computational thinking, algorithms and programming
Lesson	Array yard (lesson 5 of the unit)
Lesson ID	pf-105
Experience ID	pf-105
Plan ID	PLAN-PF-L05
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Big question, outcomes and success criteria

Big question — How does one name hold many values, and how is code broken into reusable pieces?

Prerequisites

The lesson before this one is pf-104. Pupils should already have finished it: the starter below retrieves from it.

Earlier in the unit: pf-101, pf-102, pf-103.

Assessable outcomes

1. **pf-105-o1** — know how one- and two-dimensional arrays are declared, read and changed
2. **pf-105-o2** — understand why arrays are zero-indexed and how a loop iterates over one
3. **pf-105-o3** — know the difference between a procedure and a function, and between local and global variables

Vocabulary, resources and preparation

Key vocabulary — Zero-indexing, Sub-program

Resources

SLIDES	presentations/PF_L05_ArrayYard_Lesson.pptx
WORKSHEET	worksheets/PF_L05_ArrayYard_Worksheet.docx
ONLINE EXPERIENCE	experience.html?id=pf-105

Before the lesson

1. Sign in on the **teacher dashboard** (teacher.html) and open **Class logins**. Enter the class or group and the usernames, one per line, then print the login cards. A pupil cannot sign in without one, so do this before the lesson rather than in it.
2. Print the worksheet, one per pupil.
3. Open the slides and the experience once yourself, on the machine you will be teaching from, so the 360 image is already cached.
4. Check headsets or devices are charged and on the school network.

How pupils get in

1. Go to `topics.html` on `revise360.co.uk`.

2. Sign in with the **username and PIN from the login card**. The class is already set against the card.
3. Choose the topic, then this lesson's card.

Worth knowing before you promise anything. In this build APP_CONFIG.backendUrl is empty, so logins, login cards and progress are stored on the device the pupil used. The teacher dashboard says so too. A pupil who moves to a different machine starts again, so keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels.

Teaching sequence

Sixty minutes including transitions. The online block is the one to move if your lesson is shorter: see the adaptation at the end.

MIN	PHASE	TEACHER	PUPILS	EVIDENCE TO CHECK
0-5	Retrieve	Starter on the board or the worksheet, from the last lesson and this lesson's prerequisites.	Answer the starter on the worksheet.	The starter box on PF_L05_ArrayYard_Worksheet.docx is filled in.
5-8	Review the starter	Take answers; correct the one most got wrong.	Correct their own starter.	A correction written, not just a tick.
8-18	Explain and model	The new idea in two or three steps, with the example worked in front of the class.	Listen and answer questions about the example.	They can say why, not only what.
18-21	Set up and settle	Sign-in, launch, and what to write before anyone taps an answer.	Sign in, open the lesson, find One-dimensional arrays.	Everyone is in and on station 1.
21-43	Online: work the stations in order	Pupils read each wall, write the key fact and the challenge, then answer at the badge.	Read each wall, write the key fact and the challenge, then answer at the badge.	The worksheet has writing on it for each of the 6 stations before the badge is tapped.
43-48	Check understanding	The hinge question, hands down.	Answer the hinge question, hands down.	The split across the options.
48-56	Apply	Exam practice on the worksheet, in silence.	Exam practice, in silence.	worksheet, Exam practice a) to c)
56-60	Exit and set up the next lesson	The exit question and the retrieval link to next lesson.	Write the exit answer.	One sentence that answers the big question.

The explanation and the worked example

This is what the walls of the experience say, station by station. Teach the first two or three from the front before anyone puts a headset on; the rest the pupils read for themselves.

1. One-dimensional arrays

- An array stores many values of the same data type under a single name, instead of a separate variable for each one.
- In the OCR Exam Reference Language, array crate[5] declares an array with five elements.
- An array is given its size when it is declared and keeps it; a list in Python can grow and shrink as the program runs.

Challenge on this wall: Explain why one array is better than five separate variables for storing five crate weights.

2. Index positions

- Each element has an index, which is its position in the array. OCR arrays begin at index 0, which is called zero-indexing.

- So the first element of array `crate[5]` is `crate[0]` and the last is `crate[4]`. Asking for `crate[5]` goes past the end and causes an error.
- An element is changed by naming its index: `crate[2] = 7` replaces whatever was already in position 2. A count-controlled loop iterates over the whole array.

Challenge on this wall: An array holds eight marks. Give the index of the first mark and the index of the last.

3. Two-dimensional arrays

- A two-dimensional array is a table of values with rows and columns. In the OCR Exam Reference Language, array `bay[3,4]` declares 3 rows of 4.
- An element needs two indices, the row first and then the column, so `bay[0,0]` is the top left corner.
- Two nested loops iterate over it: the outer loop moves down the rows and the inner loop across the columns.

Challenge on this wall: A noughts and crosses board is stored as array `board[3,3]`. Give the two indices of the centre square.

4. Procedures and functions

- A sub-program is a named block of code that the main program can call, as many times as it needs, without the code being written out again.
- A procedure carries out a task but returns no value. In the OCR Exam Reference Language it runs from procedure to endprocedure.
- A function always returns a value to the line that called it, using `return`, and runs from function to endfunction.

Challenge on this wall: Decide whether printing a welcome banner and working out an average should each be a procedure or a function, and say why.

5. Parameters and returns

- A parameter is a variable in the sub-program's definition that receives a value when the sub-program is called.
- Parameters let the same sub-program work on different data every time, so it does not have to be rewritten.
- `return` sends one value back to the calling line and ends the function straight away.

Challenge on this wall: Write a function in the OCR Exam Reference Language that takes two marks as parameters and returns their average.

6. Local and global

- A local variable is declared inside a sub-program and only exists while that sub-program is running. Nothing outside it can see the variable.
- A global variable can be used anywhere in the program. In the OCR Exam Reference Language it is declared with the keyword `global`.
- Local variables are safer, because two sub-programs can use the same name without clashing and the memory is freed when the sub-program ends.

Challenge on this wall: Give one advantage and one disadvantage of making a player's score a global variable.

Retrieval and the check for understanding

Starter — A program records the number of items in each of five crates. Write down why an array is a better choice than five separate variables, and write the line that would declare it.

Answer space: 3 lines on the worksheet.

The answer is in the teacher answers for this unit, not on this plan: a plan left on a desk should not give it away.

Hinge question

Use this one, hands down, before anyone moves on to independent work. It is a real activity from the experience (pf-105-s4-c1), so the wording a pupil sees matches the wording you say.

What is the difference between a procedure and a function?

Correct: A function returns a value to the code that called it, and a procedure does not

What a pupil might choose instead

- A procedure may be called many times, and a function only once
- A function must be written inside a loop, and a procedure must not
- A procedure may be given parameters, and a function may not

Count the hands on each. If more than a few of the class are on one wrong option, that option is the misconception to teach against before anyone goes on: go back to the station this question sits on and work through its example again, aloud, before the independent work.

The course's own response to a wrong answer here: Both are sub-programs and both can take parameters. The return value is the only difference.

One response, not four. one response covers every wrong option; it does not say what to revisit for each one, so decide your own reteach for each option before the lesson - the list above is the start of it.

Guided to independent practice

STAGE	ACTIVITIES	WHAT IT IS
Recognise	13	The answer is on the screen and the pupil picks it: multiple choice, choosing several, matching a pair.
Construct	1	Nothing is offered. The pupil classifies, orders, fills in a table or works a number out, and types or places the answer.

Scaffolds already in the product, and how they fade

- **Hint** is a ladder, not a door: the first rung names the idea, the next shows the shape of the code, the next how it starts, the last animates the technique on different data. A pupil asks for each rung. None of them is the answer.
- **Run** is free and unmarked; **Check my answer** marks. A pupil may run as often as they like, which is what makes trial and error safe.
- **I need help** says to ask you, and says plainly that the question still has to be completed. It does not notify anyone and it does not unlock anything.
- The worked example appears on every early activity and on none of the late ones. That fade is deliberate; it is not a missing example.

Accessibility

- **Read aloud** on every task window reads the task and its brief, with the prescribed values read as values rather than as punctuation.
- Text is 16px or larger throughout, and a question window scrolls rather than shrinking its type.
- Prescribed data is boxed and in a monospace face as well as coloured, so it still reads as code in black and white, in a high-contrast mode, and to a pupil who does not separate green from amber.
- **Lower literacy and EAL:** pre-teach the vocabulary above; let the pupil say the key fact before writing it; accept the key fact in two or three words.
- **SEND:** the sequence is compulsory but the pace is not. A pupil may finish a station next lesson; the progress saves after every answer and reopens at their earliest unfinished question.
- A headset is never required. Every experience works on a laptop or a tablet with a mouse or a finger.

Delivering the online experience

Before anyone enters

- Worksheet out, pen in hand. The rule is write first, then answer at the badge: the first answer is the one that counts.
- Say where the stations are. They are numbered around the room: 1 and 2 to the right, 3 and 4 behind, 5 and 6 to the left, and the final challenge on the floor.
- Say that a badge marked **Not yet** is not a fault. It opens when the station they are on is finished.

The stations, and what to record at each

#	STATION	ACTIVITIES	RECORD ON PAPER
1	One-dimensional arrays	2	An array is...

#	STATION	ACTIVITIES	RECORD ON PAPER
2	Index positions	2	Zero-indexing means...
3	Two-dimensional arrays	2	A two-dimensional array is...
4	Procedures and functions	2	A procedure... A function...
5	Parameters and returns	2	A parameter is...
6	Local and global	2	A local variable... A global variable...
final	Final challenge: total the crates	2	the answer to the station's challenge

Where to pause

Stop the class after station 2 and take answers to its challenge out loud. That is the point where a pupil who has misread the technique can still be put right cheaply.

Afterwards

Headsets off and down. Ask three pupils for the key fact from a station of your choosing, in their own words and not read off the sheet. Then the exit question.

Assessment, evidence and unfinished work

ID	QUESTION	MARKS	WHERE
pf-105-x1	An array is declared as <code>array scores[6]</code> . State the index of the first element and the index of the last element.	2	worksheet, Exam practice a); deck, Exam practice slide
pf-105-x2	Explain the difference between a procedure and a function.	3	worksheet, Exam practice b); deck, Exam practice slide
pf-105-x3	A program stores five crate weights in array <code>crate[5]</code> . Write an algorithm that outputs the total of all five weights, using the OCR Exam Reference Language or a language you know.	5	worksheet, Exam practice c); deck, Exam practice slide

Expected answers

Model answers and marking guidance are teacher-only and are not printed on this plan. They are in the unit's teacher guide pack, and the per-activity marking for this lesson is in `answers/records/pf-105.json`, which is not published with the website.

Common errors

The wrong options the activities offer are the errors this lesson expects. The commonest to watch for:

- **In the OCR Exam Reference Language, what does `array crate[5]` declare** — An array with six elements; A variable holding the number 5
- **What is the main advantage of using one array instead of ten separate** — An array uses no memory at all; Each value in an array can be given its own name
- **Using zero-indexing, what is the index of the last element of array `c`** — 5; 1
- **`crate[2]` holds 31. What does `crate[2] = 7` do?** — It adds 7 to the end of the array; It adds 7 to 31, giving 38
- **In the OCR Exam Reference Language, what does `array bay[3,4]` declare** — A one-dimensional array of 34 elements; Two arrays, one with 3 elements and one with 4

Unfinished work

Record where each pupil stopped, not a tick. They resume at their earliest unfinished question, and a station left half-done is the thing to open next lesson. Do not mark the lesson complete for a pupil who did not finish it: the score screen already shows what was and was not done, and that is the honest record.

Exit, next lesson and homework

Exit check

One of these, written, not a show of hands:

- How a one- and a two-dimensional array are declared and used
- Why arrays start at index 0, and how to loop over one
- Procedures, functions, parameters, return values and local versus global variables

The confidence columns on the worksheet are a supplement to this, not a substitute: a pupil who rates themselves confident and cannot answer the exit question has told you nothing.

Retrieval link to next lesson

Next is pf-106, Chance machine. Use this lesson's big question as next lesson's starter, so the first thing they do is retrieve it.

Homework

Finish any station left open, on the website, from the question they stopped at. If the class has no devices at home, set the exam practice on the worksheet instead and mark it next lesson.

Shorter lessons, longer lessons and no internet

A 40 to 45 minute lesson

Keep the starter, the explanation and the model at full length: cutting those is what makes the online work fail. Take the time out of the online block and the apply phase. Pupils stop where they are; their progress saves after every answer and reopens at their earliest unfinished question, so the next lesson starts there.

If the internet or the devices are unavailable

Teach from the slides and the worksheet. The slides carry the same teaching as the station walls, and the worksheet carries the key facts, the challenges and the exam practice. Set the online stations when the devices are back.

There is no offline version of the experience, and this plan does not contain one. Do not hand out a printed walkthrough of the stations as a substitute: the experiences are used on the website.

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